Nafs-Aql Development Roadmap

Version 2.2 (20 October 2026): Phases 0–6 implemented, Phases 7–9 added, road ahead

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Abstract

This roadmap takes Nafs-Aql from the current draft to a production-ready release in seven phases. Each phase has objectives, scope, deliverables, work packages with task identifiers, dependencies, measurable entry and exit criteria, and the requirement identifiers from the Technical Specification that it satisfies. Milestones are decision gates with explicit go/no-go criteria and fallbacks. The minimum viable product is defined with acceptance tests. Effort, duration and compute figures are *estimates* under the stated assumptions. The backlog is exported as `backlog.csv` and `roadmap.md`; the task and requirement tables are generated from the same source files as the specification. Version 2.0 records where the implementation stands, adds Phase 7 (CPU inference performance) and sets out the road ahead in Section 19.

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1 Overview

1.1 Phases and timeline

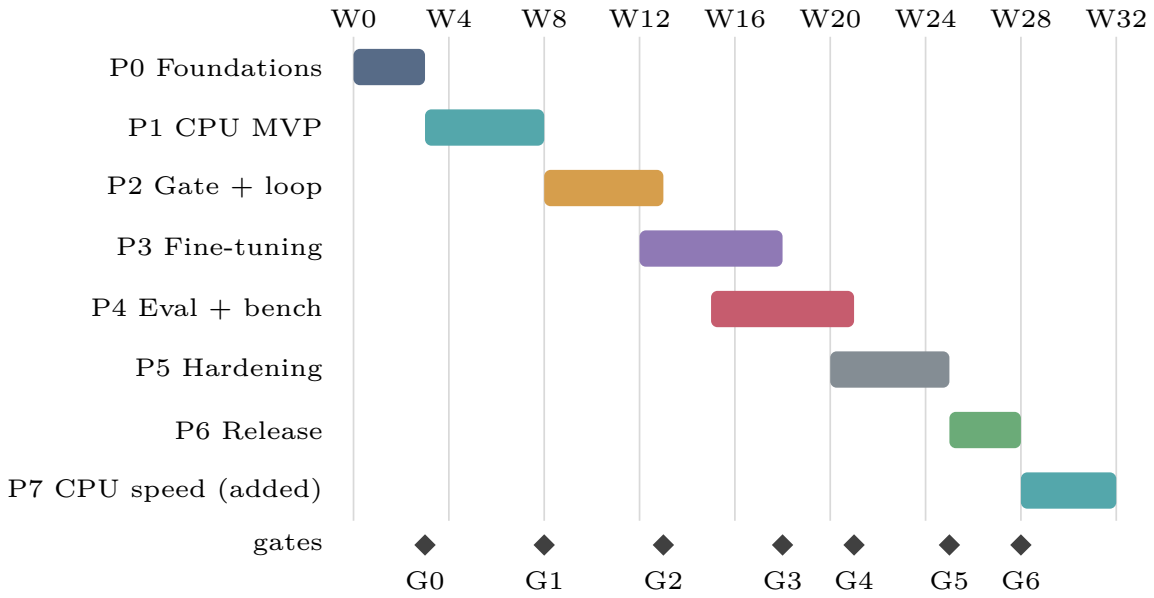


Figure 1: Timeline (weeks from project start; estimates). Diamonds are decision gates G0–G6.

Phases overlap where their work packages are independent (for example, fine-tuning data work starts while the gate is being calibrated). Total estimated effort is 69.8 person-weeks over 32 calendar weeks (28 for Phases 0–6, plus the added Phase 7) (54 tasks; totals by phase: P0: 7.5; P1: 13.0; P2: 10.5; P3: 5.5; P4: 6.5; P5: 6.5; P6: 4.5; P7: 9.5; P8: 2.2; P9: 4.0). The critical path (Figure 2) runs through serving, the measurement baseline, the calibrated gate, the data pipeline, training, promotion, hardening and the release gate.

1.2 Assumptions behind the estimates

1. Team of three to four engineers (Section 14) working concurrently; one GPU day per week available from Phase 3.
2. The reference CPU machine is available full time for baselines and tests; evaluation runs of 4 600 tasks on the slow path take about two to three machine-days and are scheduled overnight.
3. Qwen3 Profile A is the default; Profile C is adopted only through gate G0 or G5.
4. No external dependency on model releases: pinned revisions are not changed mid-programme (any change re-runs the regression suite and, if the gate's calibration is affected, the Phase-0 baseline).

2 Phase 0: Foundations and Feasibility (weeks 0–3)

Objectives. Serve both models on the reference machine; establish the measurement baseline that every later parameter depends on; create the workspace and CI. **Scope.** Serving recipes, model manifest, minimal harness with exact verifiers, 4 600-task baseline run, accuracy-versus-budget curve, hybrid-model benchmark, Cargo workspace. **Deliverables.** Serving scripts (llama-server, vLLM-CPU container), `models/manifest.json`, harness v0, Phase-0 report with q, (r, φ), $e_1(s)$, $e_2(s)$, cost ratio, discordance, Beta fits, budget curves, llama-bench table. **Requirements satisfied.** FR-050, NFR-012, NFR-023, MR-001, MR-003, MR-005, OR-001, ER-001, ER-010, ER-011.

ID	Epic	Task	PW	Depends	MoSCoW
P0-T01	E0.1 Environment	Build llama.cpp and vLLM-CPU container; publish serving recipes for Profile A	1.0	–	Must
P0-T02	E0.1 Environment	Pin model revisions, GGUF hashes, chat-template hashes; write model manifest	0.5	P0-T01	Must
P0-T03	E0.2 Measurement	Implement minimal Eval Harness (dataset loaders, exact verifiers for GSM8K/HumanEval/schema, metrics)	2.0	P0-T01	Must
P0-T04	E0.2 Measurement	Run fast-only and slow-only baselines on $\geq$ 4600 tasks; fit q, (r,phi), e1(s), e2(s), cost ratio, discordance	1.5	P0-T03	Must
P0-T05	E0.2 Measurement	Measure accuracy vs thinking budget per domain (H4) and acceptance-rate distribution	1.0	P0-T04	Must
P0-T06	E0.3 Hybrid gate	llama-bench Qwen3.5-2B/4B vs Qwen3-1.7B/4B on target CPU (H7); decide Profile C	0.5	P0-T01	Must
P0-T07	E0.4 Workspace	Create Cargo workspace, CI (fmt, clippy, test on Linux and macOS), config schema	1.0	–	Must

Dependencies. None external; P0-T03 depends on P0-T01. **Entry criteria.** Reference machine and GPU credit available; draft and report accepted. **Exit criteria (measurable).** (1) Both Profile-A servers answer the OpenAI-compatible probe with log-probabilities and separated reasoning; (2) baseline run of $\geq$ 4 600 tasks across $\geq$ 5 domains completed with per-domain tables; (3) exact verifiers for GSM8K, HumanEval and schema tasks have $\hat{\varphi}$ upper bound ≤ 0.01 on ≥ 200 mutants; (4) llama-bench table for Qwen3-1.7B/4B and Qwen3.5-2B/4B recorded; (5) CI green on Linux and macOS.

3 Phase 1: CPU-Only MVP (weeks 3–8)

Objectives. The smallest build that demonstrates the idea end to end on the reference machine. **Scope.** Gateway, backends, orchestrator with a fixed-rule gate, verifier hub v1 (code sandbox, schema), memory, chat UI with reasoning-mode and assurance badges, MVP acceptance tests. **Deliverables.** Running system, OpenAPI document served, acceptance test report with measured latencies. **Requirements satisfied.** FR-001, FR-002, FR-003, FR-005, FR-006, FR-012, FR-013, FR-023, FR-024, FR-026, FR-030, FR-031, FR-035, FR-060, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005, NFR-006, NFR-011, NFR-021, NFR-022, IF-001, IF-002, IF-003, IF-004, IF-005, IF-006, IF-007, IF-008, IF-010, MR-002, SR-002, OR-006, ER-006.

ID	Epic	Task	PW	Depends	MoSCoW
P1-T01	E1.1 Gateway	axum gateway: /v1/chat, /v1/chat/stream (SSE), auth, rate limit, metrics, OpenAPI at /openapi.json	2.0	P0-T07	Must
P1-T02	E1.2 Backends	Backend trait + llama-server and vLLM implementations: logprobs, parallel samples, thinking toggle, retry, semaphore	2.0	P0-T07	Must
P1-T03	E1.3 Orchestrator	Trace-state automaton with fixed-rule gate, budget forcing, caps, decision record, assurance levels	2.5	P1-T02	Must
P1-T04	E1.4 Verifiers	Verifier Hub v1: code sandbox (subprocess limits), JSON Schema verifier	2.0	P0-T07	Must
P1-T05	E1.5 Memory	SQLite conversation store, JSONL traces, context cap and prompt-cache prefix preservation	1.5	P0-T07	Must
P1-T06	E1.6 Web UI	Chat room UI (askama + htmx + WebSocket), reasoning mode and assurance badges, controller event stream	2.0	P1-T01	Must
P1-T07	E1.7 MVP acceptance	Profile selection, MVP acceptance tests, latency targets measured on reference machine	1.0	P1-T03, P1-T04, P1-T05, P1-T06	Must

Dependencies. P0-T07 (workspace), P0-T01 (serving). **Entry criteria.** Phase-0 exit criteria (1), (3), (5). **Exit criteria (measurable).** MVP acceptance tests of Section 13 pass; fast-path TTFT $\leq 2\,\mathrm{s}$ p95 and 150-token reply $\leq 12\,\mathrm{s}$ p95 on the reference machine over 100 requests; slow-path reply with a 1 024-token budget $\leq 240\,\mathrm{s}$ p95 over 30 requests; every automaton transition covered by a test; coverage of `nafs-core` $\geq 80\%$.

4 Phase 2: Controller, Gate and Verification Loop (weeks 8–13)

Objectives. Replace the fixed rule by the calibrated gate of the formal model; add the verified retry loop, the audit stream and the ledger; prove non-inferiority at three points. **Scope.** Feature extraction and calibration, cost-ratio rule and budget mode, sequential verified sampling with caps from the Beta fit, evidence-fed retries, math verifier and mutation harness, judge hard gate, audit stream and capability table, resilience and injection delimiting, gate-G2 evaluation. **Requirements satisfied.** FR-004, FR-010, FR-011, FR-014, FR-015, FR-020, FR-021, FR-022, FR-025, FR-032, FR-034, FR-040, FR-041, NFR-010, NFR-020, MR-004, SR-001, SR-003, ER-002, ER-003, ER-004, ER-005.

ID	Epic	Task	PW	Depends	MoSCoW
P2-T01	E2.1 Calibrated gate	Feature extraction, logistic/isotonic estimator, cost-ratio rule, budget mode, recalibration job	2.5	P0-T04, P1-T03	Must
P2-T02	E2.2 Verified loop	Sequential verified sampling, evidence-fed retries, caps from Beta fit, R≤1 on non-verifiable	2.0	P1-T03, P0-T05	Must
P2-T03	E2.3 Verifier quality	Math verifier (numeric + SymPy sidecar), mutation-testing harness, (r,phi) publication, judge hard gate	2.0	P1-T04	Must
P2-T04	E2.4 Audit and ledger	5 percent audit stream, capability table with unmeasured labels, adapter-hash keying	1.5	P2-T01	Must
P2-T05	E2.5 Resilience	Backend reconnect, degraded mode, prompt-injection delimiting	1.0	P1-T02	Must
P2-T06	E2.6 Gate G2 evaluation	Coupling test ($\geq$ 200 trials), calibration test ($\geq$ 2900), 3-point non-inferiority ($\geq$ 700 pairs)	1.5	P2-T01, P2-T02, P2-T03	Must

Dependencies. P0-T04/T05 (baseline, budget curve), P1-T03 (orchestrator), P1-T04 (verifiers). **Entry criteria.** Phase-1 exit; Phase-0 tables available. **Exit criteria (measurable).** ECE of $\hat{e}_1 \leq 0.05$ on $\geq 2\,900$ held-out samples; routing accuracy ≥ 0.70 at $g \leq 0.30$ (H1); cascade non-inferior to slow-only within 3 points on ≥ 700 pairs (ER-002) at cost ratio ≤ 0.60 (H2); coupling test: adoption upper bound $\leq 5\%$ on ≥ 200 trials (H5); every verifier publishes (r, φ) with intervals; degraded mode demonstrated by killing the slow backend during a load test with zero lost traces.

5 Phase 3: Fine-Tuning with Unsloth (weeks 12–18)

Objectives. Close the distillation loop once under the promotion gate. **Scope.** Dataset builder with privacy redaction, synthetic data, Unsloth LoRA scripts and GPU runbook, GGUF export, promotion gate, regression suite, rollback. **Requirements satisfied.** FR-033, FR-070, TR-001, TR-002, TR-003, TR-004, TR-005, TR-006, TR-007, TR-008, SR-004, OR-005, ER-008.

ID	Epic	Task	PW	Depends	MoSCoW
P3-T01	E3.1 Data pipeline	Dataset builder from verified traces; schema; dedupe; domain cap; abstain and replay mixing; privacy redaction	1.5	P2-T04	Must
P3-T02	E3.2 Synthetic data	System 2 synthetic generation on benchmark-like prompts with held-out verifier	1.0	P3-T01	Should
P3-T03	E3.3 Unsloth training	Unsloth LoRA scripts, GPU runbook (T4), checkpoint and adapter hashing, GGUF export	1.5	P3-T01	Must
P3-T04	E3.4 Promotion	Promotion gate with confidence bounds, regression suite, human sign-off, one-command rollback	1.5	P3-T03	Must

Dependencies. P2-T04 (ledger, audited traces), GPU access. **Entry criteria.** $\geq 5\,000$ verified traces or synthetic examples available; Phase-2 exit. **Exit criteria (measurable).** One adapter trained on a GPU with recorded versions and hashes; merged GGUF loads and passes the regression suite; promotion gate evaluated with confidence bounds on a held-out verifier (H6), whatever the

verdict; rollback demonstrated in under one minute; privacy redaction tested on a seeded set with zero leaks.

6 Phase 4: Evaluation Suite and Performance Console (weeks 15–21)

Objectives. Make the system measure itself. **Scope.** Paired statistics and report export, load generator and bench API, dashboard with gate status, drift monitor. **Requirements satisfied.** FR-042, FR-051, FR-052, FR-053, NFR-007, IF-009, ER-007.

ID	Epic	Task	PW	Depends	MoSCoW
P4-T01	E4.1 Statistics	Paired tests, CIs, non-inferiority, Bonferroni in harness; report export	1.5	P2-T06	Must
P4-T02	E4.2 Performance console	goose load generator, bench runs API, latency quantiles, concurrency protocol	2.0	P1-T01	Must
P4-T03	E4.3 Dashboard	Evaluation dashboard: runs, calibration plots, routing accuracy, gate status	2.0	P4-T01	Should
P4-T04	E4.4 Drift	CUSUM drift monitor with negative control on ledger	1.0	P2-T04	Should

Dependencies. P2-T06 (evaluation data), P1-T01 (gateway). **Exit criteria (measurable).** All metrics of the specification implemented with unit tests on synthetic data; load tests at concurrency 1, 2, 4 reported with p50/p95/p99 and NFR-007 verdict; dashboard shows G0–G4 status from stored runs; CUSUM raises an alarm on an injected shift of 2σ within 50 observations and none on the negative control over 500.

7 Phase 5: Hardening and Optional GPU Scale-Out (weeks 20–25)

Objectives. Production readiness and the upgrade decisions. **Scope.** Security review and tests, TLS and secrets, packaging, deployment recipes, monitoring and alerts, vLLM CUDA profile, Profile-C evaluation, optional retrieval memory. **Requirements satisfied.** FR-061, SR-006, OR-002, OR-003, OR-004.

ID	Epic	Task	PW	Depends	MoSCoW
P5-T01	E5.1 Hardening	Security review, TLS and secrets, sandbox escape tests, packaging with manifest	2.0	P4-T02	Must
P5-T02	E5.2 Deployment	docker-compose (vLLM-CPU) and service files (llama-server); monitoring and alerts	1.5	P5-T01	Must
P5-T03	E5.3 GPU scale-out	vLLM CUDA profile with both models resident, higher N and budget, speculative decoding	1.5	P5-T02	Could
P5-T04	E5.4 Upgrade decision	Profile C (Qwen3.5) evaluation on full suite; optional retrieval memory	1.5	P0-T06, P4-T01	Could

Dependencies. P4-T02, P3-T04, P0-T06. **Exit criteria (measurable).** Zero findings of severity high or above open from the security review; sandbox escape tests pass; one-command deployment on a clean Linux host (vLLM-CPU) and on the reference machine (llama-server); alerts fire in a drill; GPU profile (if pursued) reaches slow-path p95 ≤ 20 s with N and b at least doubled; Profile-C decision recorded with data.

8 Phase 6: Release and Research Publication (weeks 25–28)

Objectives. Release and publish. **Scope.** One-point non-inferiority gate, full regression, scholar review, documentation, research write-up with measured results. **Requirements satisfied.** SR-005.

ID	Epic	Task	PW	Depends	MoSCoW
P6-T01	E6.1 Release gate	1-point non-inferiority on $\geq$ 6200 pairs; full regression; scholar review of analogy copy	2.0	P5-T01	Must
P6-T02	E6.2 Documentation	Operator and user documentation; API reference; runbooks	1.0	P6-T01	Must
P6-T03	E6.3 Publication	Research write-up with measured results; artefact release	1.5	P6-T01	Should

Exit criteria (measurable). Non-inferiority within 1 point on $\geq$ 6 200 pairs; regression suite green; scholar review sign-off recorded; documentation complete (API reference generated from OpenAPI, operator runbook, user guide); write-up submitted with all numbers traceable to stored evaluation runs.

9 Phase 7: CPU Inference Performance (weeks 28–32, added)

Objectives. Make the CPU path as fast as the hardware allows, and know where every second goes. Phase 7 was added on 18 October 2026, after Phases 0–6 had been implemented on the reference machine, in response to the goal of maximum speed on CPU with GPU work deferred. **Scope.** Per-stage latency attribution with the server's prefill, decode and cache timings; engine tuning under interleaved A/B tests; early escalation in the streaming draft; EAGLE-3 speculative heads; repacked 4-bit weight formats; the efficiency frontier in the Lab; an end-to-end sweep and documentation refresh. **Requirements satisfied.** NFR-030, NFR-031, FR-080, FR-081, ER-020, FR-082.

ID	Epic	Task	PW	Depends	MoSCoW
P7-T01	E7.1 Observability	Per-stage latency with server prefill/decode/cache split; /v1/perf/stages; inspector waterfall and Performance breakdown	1.0	P4-T02	Must
P7-T02	E7.2 Engine	Interleaved A/B harness and study of flash attention, BLAS prefill, KV-cache type and threads; verifier warm-up	1.0	P7-T01	Must
P7-T03	E7.3 Cascade	Early escalation in the streaming draft (fixed rule) with decision-agreement study	1.0	P7-T01	Should
P7-T04	E7.2 Engine	EAGLE-3 heads for Qwen3-1.7B and Qwen3-4B (convert, serve, measure acceptance and speed-up)	1.0	P7-T02	Could
P7-T05	E7.2 Engine	Weight-format study: Q4_0 / IQ4_NL with AVX2 repack kernels against Q4_K_M (speed and accuracy on regression_500)	1.0	P7-T02	Could
P7-T06	E7.4 Lab	Efficiency frontier in Lab insights: early escalation, saved tokens, routing agreement per domain	1.0	P7-T03	Should
P7-T07	E7.5 Release	End-to-end sweep, UI screenshots, technical report and slides refresh, forward roadmap	1.0	P7-T06	Must
P7-T08	E7.3 Cascade	Adaptive reasoning exit: probe-and-verify on the raw continuation path, with a token-saving study	1.0	P7-T03	Should
P7-T09	E7.3 Cascade	Judge on System 1: measure the small judge's (r, phi) with the rates harness and adopt it if the rates hold	0.5	P7-T01	Could
P7-T10	E7.6 Backends	vLLM CPU backend in Docker (carstenz/vllm-smoke image, BF16, one model in two modes) with an end-to-end check against the gateway	1.0	P5-T02	Should

Dependencies. P4-T02 (load-test protocol), P2-T01 (gate). **Exit criteria (measurable).** Every reply carries stage timings and the Performance page attributes at least 95% of wall time to named stages; every adopted serving change has an interleaved A/B record; early escalation agrees with full-draft decisions on at least 95% of a held-out prompt set or is switched off by default; EAGLE-3 or a repacked format is adopted only with a measured decode speed-up of at least 1.2× and identical greedy output. **Status (19 October 2026).** P7-T01–T03, P7-T07 and P7-T08 (adaptive reasoning exit, added) done; P7-T04 measured and not adopted for System 1 (no decode gain at about 50% acceptance on four cores; opt-in kept); P7-T05 prepared; P7-T06 partly done (policy map; saved tokens reported by the studies).

10 Phase 8: Sequential Evidence (added 20 October 2026)

Objectives. Decide, from first principles and measurements, what the neuroscience of decision making and the quantum theories of consciousness contribute to the accuracy and speed of Nafs-Aql, and implement only what carries a proof and a measurement. **Scope.** A review with explicit

pass/fail tests (`docs/research/quantum_neuro_review.md`, report Section 8); a confidence-signal study of System 1 drafts against exact labels (ROC, AUROC with bootstrap intervals, recall at the operating escalation rate); the lowest-window confidence signal as a gate feature with an exact early-escalation rule (a running minimum, so an early crossing is the full draft's decision); a posterior stopping rule with a collapsing bound for the adaptive exit on non-verifiable tasks, compared in a paired Lab run; Wald's bound on the mean-based early escalation and the optimal probe interval as validations of existing settings. **Requirements satisfied.** FR-083, FR-084, FR-085, ER-021. **Out of scope.** Anything from quantum consciousness: no mechanism passed the transferability test (Proposition Q2 of the report), and the result is recorded as negative rather than deferred.

Exit criteria. The signal study is stored with its sample size and intervals; the window rule's threshold and domains come from it; the posterior rule is on by default only if the paired run shows unchanged answers and no token increase; the review names, for every idea not adopted, the test it failed.

11 Phase 9: Rooms (added 20 October 2026)

Objectives. Make shared rooms a complete, trustworthy feature: a room can be created, renamed and deleted for good, a member always sees a consistent transcript however and whenever it joins, and the page survives a dropped connection. **Scope.** The room store (unique normalised names, a bound on the number of rooms, rename, and a deletion that purges the room's conversation and every ledger row made from it in one transaction); the room hub (a welcome snapshot carrying a history cursor and the pending questions, taken together with the subscription so that each turn is delivered exactly once; unique display names; presence, queue-state and typing frames; heartbeat; cancellation of a running answer when the room is deleted; tombstones that expire); the rebuilt Rooms page (search, unread markers, deep links, inline rename, confirmed delete, export, automatic reconnection, drafts, windowed history, a phone drawer); a live test with simulated members. **Requirements satisfied.** FR-090, FR-091, FR-092, FR-093, ER-022. **Out of scope.** Accounts and per-room permissions: the gateway has none, so anyone who can reach it can administer any room, which the user guide and runbook state plainly.

Exit criteria. The live rooms test passes, including a deletion while an answer is being written; a stress test of the join snapshot shows every turn delivered exactly once under concurrent joins; deleting a room leaves no reply, decision, memory or conversation row behind.

12 Milestones and Decision Gates

Table 1: Decision gates with go/no-go criteria and fallbacks

Gate	Week	Go criteria	No-go action
G0	3	Phase-0 exit criteria met; Profile decision: Qwen3.5 adopted only if decode $\geq$ 70% of dense Qwen3 at equal size (H7)	Keep Profile A; if the baseline cannot reach 4 600 tasks in time, release the gate for a 2 000-task interim run and schedule the remainder in Phase 2
G1	8	MVP acceptance tests pass; latency targets met	If NFR-004 fails (slow path too slow), reduce b to 512 and N to 1, or switch to Profile B; if memory exceeds 8 GB, use Q4_K_M for both models and a 4 096 context

Gate	Week	Go criteria	No-go action
G2	13	ECE $\leq$ 0.05; routing accuracy $\geq$ 0.70; 3-point non-inferiority; coupling $\leq 5\%$	If calibration fails, keep the fixed rule in budget mode (quantile threshold) and extend calibration data; if non-inferiority fails, raise λ until it holds and report the cost ratio honestly; if coupling fails, confirm draft-blindness and measure the collision term
G3	18	First adapter trained and gate evaluated; rollback demonstrated	If no GPU access, defer Phase 3 and continue collecting verified traces; if the gate fails, do not promote, document, and run one more cycle with more data
G4	21	Evaluation suite complete; load tests reported	If NFR-007 fails, document the single-session profile and make concurrency a GPU-profile feature
G5	25	Security review clean; deployment recipes verified; GPU/Profile-C decisions recorded	If findings remain, release is blocked until fixed; GPU and Profile C are optional and may be dropped without affecting release
G6	28	1-point non-inferiority; regression green; scholar sign-off	Release with the 3-point result and a documented limitation if 6 200 pairs are not reachable; scholar review is blocking for the analogy text only

13 MVP Definition

The MVP is Phase-1 output: Profile A on the reference machine, llama-server backends, gateway with `/v1/chat`, `/v1/chat/stream` and chat rooms, the fixed-rule gate with the forced-escalation rule for failed verification, the code and schema verifiers, draft-blind slow path with budget forcing, assurance levels, decision records and traces, and the chat UI. **Acceptance tests.**

1. **MVP-A1 Fast reply.** A greeting returns a `fast/unverified` reply with TTFT ≤ 2 s.
2. **MVP-A2 Verified code.** A HumanEval-style task with tests returns `verified` (fast or slow) and the trace shows the verifier pass; a task whose draft fails tests is escalated (rule `forced_failed_verification`).
3. **MVP-A3 Slow path.** A GSM8K item with the gate forced to escalate returns a reply within 240 s with `reasoning_mode=slow` and a usage block showing reasoning tokens $\leq b$.
4. **MVP-A4 Abstention.** With $b = 64$ the system returns `abstained` with an uncertainty notice rather than an empty answer.
5. **MVP-A5 Bounded loop.** With $N = 2$, $R = 2$ a never-passing verifier produces at most $1 + N(R + 1) = 7$ model calls.
6. **MVP-A6 Streaming.** The SSE stream carries `decision`, `token`, `verification` and `done` events in order.
7. **MVP-A7 Rooms.** Two browser sessions in one room see each other's messages and badges.
8. **MVP-A8 Records.** Every reply's decision record and trace are retrievable and validate against the JSON Schemas.
9. **MVP-A9 Degradation.** Killing the slow backend yields a `degraded` event and a fast reply, not an error.

Table 2: Team and effort estimates (person-weeks, PW)

Role	Responsibilities	PW	Notes
Rust engineer (lead)	gateway, orchestrator, backends, memory, UI	20	full time, weeks 0–28
ML / evaluation engineer	harness, calibration, statistics, ledger, training pipeline	18	full time, weeks 0–28
Rust engineer (second)	verifiers, sandbox, bench, dashboard, security	12	weeks 3–25
Research lead (part time)	formal model, Phase-0 analysis, publication, scholar liaison	4	20% over the programme
Total		69.8	matches the task estimates by phase

14 Effort and Resources (Estimates)

Compute. CPU: the reference machine (or a 16-core AVX-512 Linux server for faster baselines; a 4 600-task slow-path run at about 150 s per item is 190 machine-hours on the reference machine and about 25 hours on a 16-core AVX-512 host). GPU: about one GPU-day per training cycle on a 16 GB T4 class card (QLoRA on 1.7B/4B); three cycles planned; an optional 24 GB GPU for the scale-out profile in Phase 5. Storage: under 50 GB (models, traces, datasets).

15 Risk Register

Table 3: Risk register (L = likelihood, I = impact; H/M/L)

Risk	L	I	Early-warning signal	Mitigation	Owner
Slow path too slow on CPU for any interactive use	H	M	NFR-004 p95 > 240 s in P1	reduce b, N; Profile B; GPU profile; honest UI expectation	Rust lead
Gate calibration fails (ECE > 0.05)	M	H	Phase-0 features uninformative (AUC < 0.65)	add features (verifier outcome, k-sample agreement, length); budget mode; conformal thresholds	ML engineer
Verifier false accepts (code tests gameable)	M	H	$\hat{\varphi}$ upper bound > 0.01 on mutants	hidden tests, mutation testing, stricter extraction; never train on accepted-but-unaudited outputs	Rust 2
Correlated errors S1/S2/judge (shared family)	H	M	coupling test adoption > 5%; judge (r, φ) poor	draft-blindness; exact verifiers carry the safety case; consensus never upgraded	Research lead
Hybrid Qwen3.5 slower than expected on AVX2	M	L	ER-011 ratio < 0.7	stay on Profile A	ML engineer
No GPU access for Phase 3	M	M	credit not secured by week 10	defer training; keep collecting verified traces; prompt-level improvements	Research lead

Risk	L	I	Early-warning signal	Mitigation	Owner
Model or runtime version drift	H	M	upstream release of Qwen3.x small models or llama.cpp behaviour change	pinned revisions and hashes; regression suite on any change	Rust lead
Sample sizes not reached in time	M	M	baseline throughput below plan by week 2	interim 2 000-task run; AVX-512 server; prioritise domains	ML engineer
Prompt injection via evidence or retrieval	M	H	injection probes succeed in CI	delimiting, sanitisation, no parameter changes from model output	Rust 2
Religious-content misrepresentation	L	H	scholar review raises objections	analogy-only wording; review before release; no authority claims	Research lead
Unsloth or transformers API churn	M	L	training skeleton fails on a new version	pin versions; dry-run stage before each cycle	ML engineer

16 Dependency and Critical-Path View

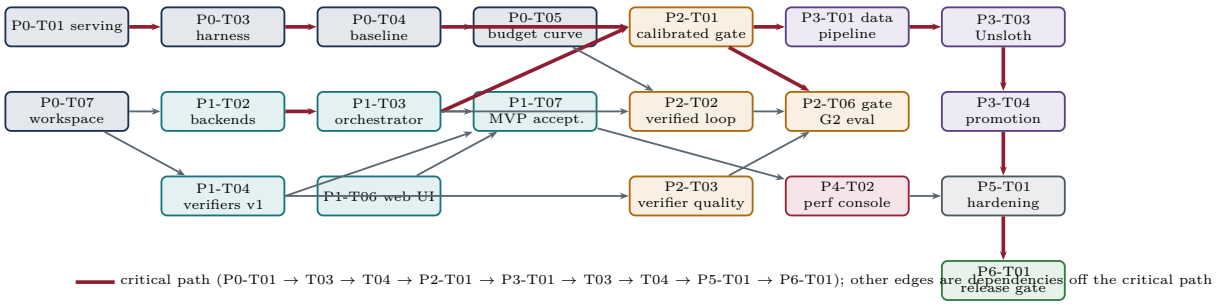


Figure 2: Dependency graph of the main tasks with the critical path highlighted.

The critical path is P0-T01 → P0-T03 → P0-T04 → P2-T01 → P3-T01 → P3-T03 → P3-T04 → P5-T01 → P6-T01: the measurement baseline gates the calibrated gate, which gates the quality of the traces that feed training, which gates promotion, hardening and release. Work off the critical path (UI, bench console, dashboard, GPU profile) can absorb slack. Phase 1 overlaps Phase 0's baseline run because the baseline is machine-bound, not people-bound.

17 Prioritised Backlog

The full backlog (54 tasks with MoSCoW priority, estimates, dependencies and requirement IDs) is exported to `backlog.csv` and `roadmap.md`. **Must** items are the critical path and the MVP; **Should** items are the judge hard gate, synthetic data, dashboard, drift monitor, replay mixing; **Could** items are budget mode, concurrency target, GPU scale-out, Profile C and retrieval memory; **Won't** (this release): multimodal input, autonomous tool use beyond verifiers, diffusion decoding.

Table 4: Hypotheses from the formal model and their validation

ID	Hypothesis (from result)	Experiment	Phase	Engineering effect if false
H1	Calibrated gate reaches routing accuracy ≥ 0.70 at $g \leq 0.30$ (Thm. 2, Prop. 1)	labelled held-out set; oracle decisions; ECE and routing accuracy	2	richer features or budget mode; raise λ
H2	Cascade costs $\leq 60\%$ of slow-only at matched accuracy (Prop. 10)	paired runs over λ grid; cost–accuracy curve	2	report honest frontier; adjust default λ
H3	Per-task acceptance rates fit a Beta per domain (Prop. 3)	goodness-of-fit on Phase-0 draws; compare predicted vs observed coverage	2	set N, R from empirical coverage curve instead
H4	Slow-path accuracy is concave in budget b (Prop. 9)	accuracy at $b \in \{128, 256, 512, 1024, 2048\}$ per domain	0	pick b by grid search, not first-order condition
H5	Draft-blind slow path adopts a wrong draft $< 5\%$ (Prop. 4)	≥ 200 trials with seeded wrong drafts, both modes	2	keep draft-blind; investigate collision term
H6	One distillation cycle gives LCB gain > 0 without regression (Prop. 7)	held-out verifier never used for collection; paired gate	3	more data, synthetic data, or no promotion
H7	Qwen3.5 hybrid decodes at $\geq 70\%$ of dense Qwen3 speed on the target CPU (Prop. 8)	llama-bench, 4 threads, pp512/tg128	0	stay on Profile A

18 Research-to-Engineering Plan

Additional checks scheduled: the self-refinement fixed point of Theorem 4 (measure u, d on a sample to confirm that unverified refinement is not monotone), the plurality floor of Theorem 6 (fraction of tasks with a wrong modal answer), and the audit-based self-model of Proposition 5 (compare audited against naive estimates).

19 Where the Programme Stands and the Road Ahead

Status on 18 October 2026. Phases 0–6 are implemented on the CPU reference machine (GPU training and GPU scale-out deferred). Gate G5 passes. G1 passed earlier in the day but failed its latest run on NFR-004: the deliberate path took 354 s against 240 s while the host was shared. G4 fails on NFR-007 (concurrency on a shared four-core host, with the documented fallback). G0, G2, G3 and G6 are pending on release-scale sample sizes, the first GPU training cycle and the scholar review. Profile C was evaluated and not adopted (more accurate, 3.7× slower). Latency attribution shows that System 1 decode speed dominates the fast path.

20 Success Metrics

- **Phase 0:** baseline tables delivered; verifier $\hat{\varphi} \leq 0.01$; CI green.
- **Phase 1:** MVP acceptance 9/9; NFR-002–NFR-005 met on the reference machine.
- **Phase 2:** ECE ≤ 0.05; routing accuracy ≥ 0.70; 3-point non-inferiority; cost ratio ≤ 0.60; coupling $\leq 5\%$.
- **Phase 3:** one gated training cycle completed; rollback < 1 minute; zero privacy leaks on the seeded test.

Table 5: The road ahead (estimates; each phase keeps the gate discipline of Phases 0–7)

Phase	Scope	Exit criteria
8 Evidence at scale	An idle or faster CPU host, or nightly slices: the 4 600-task baseline, 700 paired items for G2, 2 900 labelled drafts, 6 200 pairs for G6; promote the calibrated gate and budget mode	G0, G2 and G6 closed with stored runs; ECE ≤ 0.05; routing accuracy ≥ 0.70
9 Tazkiya cycle	5 000 verified traces (live and synthetic); the first Unsloth LoRA on a rented GPU; promotion under TR-007; rollback drill; re-fit the EAGLE-3 head to the distilled System 1	G3 closed; LCB gain > 0 without regression
10 Engine	EAGLE-3 or MTP speculation for both models; repacked 4-bit formats; adaptive early exit from System 2 reasoning; a router learned from the policy map and Both-trial preferences	decode speed-up $\geq 1.2\times$ with identical greedy output; no accuracy loss on regression_500
11 Product and review	Scholar review of the analogy text (SR-005); GPU profile for interactive deliberation; multilingual interface (Indonesian, Arabic); retrieval-grounded checks for open-domain answers	SR-005 signed off; slow-path p95 ≤ 20 s on the GPU profile

- **Phase 4:** full metric suite with tests; load reports at three concurrency levels; drift alarm sensitivity and specificity as specified.
- **Phase 5:** no open high-severity findings; clean-host deployment; decisions on GPU and Profile C recorded.
- **Phase 6:** 1-point non-inferiority; documentation and publication delivered.
- **Phase 7:** stage timings on every reply; interleaved A/B records for every serving change; early-escalation agreement $\geq 95\%$; speculative heads adopted only with a measured speed-up.
- **Phase 8:** a stored ROC study decides the gate's confidence signal; the window rule's early decisions equal full-draft decisions by construction (unit test of the running-minimum property); the posterior exit rule is default only after a paired Lab run; quantum-consciousness ideas are recorded with the test they failed.
- **Programme:** a CPU-only dual-process assistant whose answers carry honest assurance levels, whose gate is derived and calibrated, whose precision on verifiable tasks is bounded by measured verifier rates, and whose fast path improves through a gated distillation loop, with every claim traceable to a stored evaluation run.